

RepGhostFAS: Lightweight and Efficient Face Anti-Spoofing for Internet-of-Things (IoT) and Edge Devices

Taha Hasan Masood Siddique, *Graduate Student Member, IEEE*, Shujaat Khan, Muhammad Shahid Jabbar, Akash Kumar, *Senior Member, IEEE*, and Kejie Huang, *Senior Member, IEEE*

Abstract—Face Anti-Spoofing (FAS) or Face Presentation Attack Detection (FacePAD) is essential for securing face recognition systems against presentation attacks. However, existing approaches suffer from poor generalization under domain shift conditions and often incur high computational costs, which limit their discriminative capability and practical deployment, especially in resource-constrained edge and Internet-of-Things (IoT) environments. To address these challenges, we introduce *RepGhostFAS*, a lightweight and efficient FAS framework that leverages the RepGhost architecture to achieve strong discriminability with low computational cost, marking its first application in the FAS domain. We tackle the domain shift problem using ensemble-based feature fusion strategies that extract complementary information from multiple domains. To further reduce model complexity, we employ knowledge distillation to transfer stronger representations from an ensemble teacher model to a compact student network. Extensive experiments on eight benchmark datasets, namely Replay-Attack, Replay-Mobile, OULU-NPU, ROSE-Youtu, SiW-Mv2, SynthASpoof, SOTERIA, and UniAttackData demonstrate that *RepGhostFAS* achieves competitive or superior performance while requiring significantly lower computational cost than existing methods. Overall, the proposed framework offers a robust, cost-effective, and deployment-friendly solution for real-world FAS applications on IoT and edge devices.

Index Terms—Face Anti-Spoofing, Face Presentation Attack Detection, RepGhost, Ensemble Learning, Knowledge Distillation.

I. INTRODUCTION

IN recent years, the rapid growth of Internet-of-Things (IoT) has significantly expanded the use of biometric authentication systems [1]. These systems, such as fingerprint and iris recognition, rely on the unique physiological characteristics of the human body for verification or identification. Among

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Taha Hasan Masood Siddique and Kejie Huang are with the College of Information Science & Electronic Engineering, Zhejiang University, Hangzhou 310027, China (email: taha@zju.edu.cn; huangkejie@zju.edu.cn).

Shujaat Khan is with the Department of Computer Engineering, College of Computing and Mathematics, and SDAIA-KFUPM Joint Research Center for Artificial Intelligence, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia (e-mail: shujaat.khan@kfupm.edu.sa).

Muhammad Shahid Jabbar is with the SDAIA-KFUPM Joint Research Center for Artificial Intelligence, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia (e-mail: muhammad.jabbar@kfupm.edu.sa).

Akash Kumar is with the Chair of Embedded Systems, Faculty of Electrical Engineering and Information Technology, Ruhr-Universität Bochum, Bochum 44801, Germany (e-mail: akash.kumar@ruhr-uni-bochum.de).

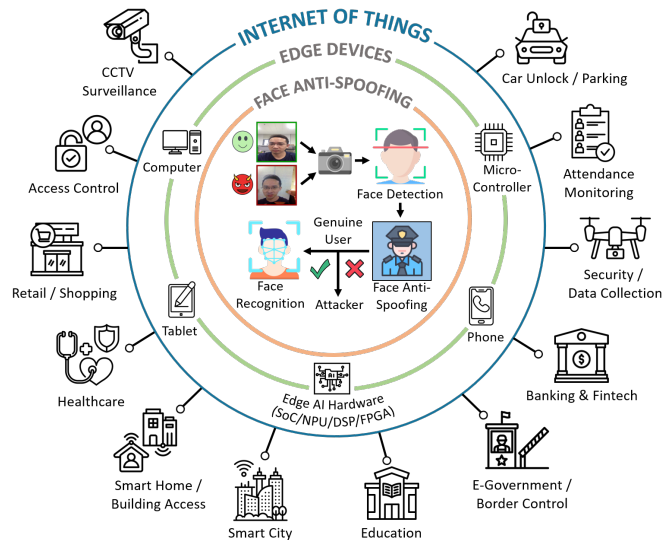


Fig. 1. Overview of the face anti-spoofing integrated into IoT and edge devices. The lightweight face anti-spoofing module performs on-device verification to prevent spoofing attacks across diverse IoT applications.

various biometric modalities, face recognition has gained particular popularity due to its contactless nature and ease of use. It is now widely employed in everyday applications, including smartphone unlocking [2], mobile payments [3], video surveillance [4], access control [5], attendance management [6], smart city systems [7], etc. However, existing face recognition systems remain vulnerable to spoofing attacks, in which attackers use artifacts such as printed photos, recorded videos, or 2D/3D masks to impersonate genuine users. This vulnerability is also reflected in real-world scenarios: (1) recent reports have shown that even age-restriction mechanisms on social media can be circumvented, with teenagers using images of their parents to bypass facial verification [8], and (2) presence of low-quality inputs caused by blur, noise, and compression can degrade the performance of face recognition and anti-spoofing systems.

Face Anti-Spoofing (FAS), also known as Face Presentation Attack Detection (FacePAD), aims to detect and prevent such spoofing attempts. Despite significant progress, existing FAS methods struggle to generalize to unseen attack types, environmental variations, and camera qualities [9]. In addition, many approaches involve high computational demands that hinder deployment on resource-constrained IoT and edge devices [10], [11]. These challenges highlight the need for a robust and lightweight FAS system capable of maintaining high accuracy while remaining practical for real-world deployment, as illustrated in Fig. 1.

A variety of FAS approaches have been proposed, ranging from hand-crafted feature methods to deep learning-based solutions [12]–[16]. Hand-crafted features heavily rely on expert design and are not learnable, making them insufficient for complex or diverse spoofing scenarios. Hybrid approaches suffer from feature incompatibility, often leading to performance saturation. Furthermore, several works have focused on image restoration and deblurring techniques [17]–[19], which can be applied to handle low-quality spoof images. While these approaches improve visual quality, they often introduce additional computational overhead and are not specifically optimized for lightweight FAS deployment. Another critical challenge is domain shift [20], where discrepancies between training and testing distributions lead to significant performance degradation. To address this, domain adaptation, domain generalization, multi-domain learning, and feature alignment methods [21]–[23] have been presented. While these methods improve robustness, they still lack generalization when facing wide variations in illumination, pose, facial appearance, and camera quality.

To build lightweight FAS models, researchers have explored pruning [24] and quantization [25] techniques to compress large models. However, these techniques tend to degrade model performance during the compression process. In addition, few studies have examined lightweight Convolutional Neural Network (CNN) architectures for FAS [26]–[29]. Knowledge distillation [30] has also emerged as an effective model compression strategy, where a high-capacity teacher model transfers its knowledge to a smaller student model. Since the teacher model typically possesses stronger representational capacity, the key idea is to encourage the student to imitate the teacher's output as closely as possible.

To the best of our knowledge, no prior study has investigated RepGhost [31] as a backbone for face anti-spoofing, nor has any work analyzed the comparison of computational complexity with different lightweight models. In this paper, we introduce *RepGhostFAS*, a lightweight and efficient FAS framework that incorporates the RepGhost backbone, enabling practical deployment on IoT and edge devices.

The main contributions of this paper are summarized as follows:

- We investigate the effectiveness of the lightweight RepGhost architecture for face anti-spoofing and present a detailed comparison of computational complexity against existing lightweight models.
- We propose three ensemble-based feature fusion strategies (concatenation, summation, and attention) to address domain shift by learning complementary cross-domain representations.
- We employ knowledge distillation to transfer knowledge from an ensemble teacher model to a single lightweight *RepGhostFAS* student model. The distilled model achieves accuracy close to the teacher while significantly outperforming baseline model with reduced computation cost.

The rest of this paper is organized as follows: Section II reviews related works. Section III presents the detailed proposed method. Section IV describes the experimental setup, and

Section V reports the experimental results. Finally, Section VI provides an in-depth discussion, and Section VII concludes the paper.

II. RELATED WORK

A. Traditional Methods

Early Face Anti-Spoofing (FAS) methods relied on hand-crafted features such as Local Binary Patterns (LBP) [32], Histograms of Oriented Gradients (HOG) [33], Scale-Invariant Feature Transform (SIFT) [34], Speeded-Up Robust Features (SURF) [35], and Image Quality Assessment (IQA) [36]. These features were typically combined with traditional classifiers such as Support Vector Machines (SVM) [37] and Linear Discriminant Analysis (LDA) [38] to differentiate genuine faces from spoofing attacks. Motion-based cues including eye-blinking [39] and lip movements [40] were also explored to detect liveness. Despite their early success, these handcrafted and motion-based methods face challenges and are easily deceived by more complex attacks such as replay attacks and high-quality mask attacks.

With the success of deep learning, Convolutional Neural Networks (CNNs) [41] and Long Short-Term Memory (LSTM) networks [42] were introduced to automatically learn discriminative features. Multimodal extensions incorporating depth [43] and reflection maps [44] further improved performance. However, deep learning approaches remain susceptible to overfitting due to limited data diversity and are sensitive to variations in illumination, pose, and camera quality. Moreover, many models require high computational and memory resources, which hinder their generalization and practical deployment on edge devices.

B. Domain Shift

In face recognition and face anti-spoofing applications, data come from diverse distributions due to variations in facial appearance, pose, illumination, and camera devices. As a result, an inherent domain shift [20] arises among different FAS datasets (e.g., Replay-Attack, Replay-Mobile, and ROSE-Youtu), leading to significant performance drops when models trained on a dataset (source domain) are tested on an another dataset (target domain). This degradation occurs because the feature distributions across domains differ substantially due to dataset-specific characteristics [45].

To address this issue, domain adaptation and domain generalization techniques [21] have been proposed to bridge the distribution gap between source and target domains. In [22], Guo et al. introduced a multi-domain learning framework that mitigates forgetting problem when adapting to new domains, while Zhang et al. [23] proposed a multi-domain feature alignment approach to achieve more generalizable feature space. However, real-world conditions remain highly diverse, making domain shift a persistent challenge. Variations in capture devices, lighting, backgrounds, and presentation attack instruments cannot be fully covered by existing datasets, limiting the effectiveness of previous approaches. This challenge motivates us to introduce ensemble-based feature fusion technique, which integrates complementary information from

multiple pre-trained models. We design three fusion strategies (concatenation, summation, and attention) to enhance feature robustness. To further balance performance and computational efficiency, knowledge distillation [30] is employed to transfer knowledge from the ensemble teacher model to a compact *RepGhostFAS* student model, improving generalization under domain shift conditions.

C. Lightweight Networks

Several studies have been presented on reducing computational complexity for FAS. Common strategies include pruning redundant filters based on FLOPs [24] and quantizing network weights or activations into low-bit representations [25]. Other approaches focus on designing inherently efficient architectures tailored for mobile and edge deployment. For example, ShuffleNet [46], [47] reduces computation through group convolutions and channel shuffling. Once-for-All [48] trains a super-network that contains multiple sub-networks with different depths, widths, kernel sizes, and resolutions, enabling flexible deployment without retraining. MobileNetV2 [49] and MobileNetV3 [50] introduced inverted residuals, linear bottlenecks, and squeeze-and-excitation modules to improve efficiency. GhostNet [51] further reduces cost by generating additional feature maps through cheap linear operations. Vision Transformers (ViT) such as MobileViTV2 [52], FastViT [53], SHViT-S1 [54], SwiftFormer [55], and TinyViT-5M [56] are less-edge friendly because self-attention introduces quadratic memory, computation growth, and relies on operations that are less efficiently optimized on low-power hardware compared to convolutional layers.

Motivated by RepGhost [31], which extends GhostNet using structural re-parameterization to improve both training flexibility and inference efficiency, we adopt it as the backbone in our *RepGhostFAS* framework. This design provides a strong balance between accuracy and computational efficiency, making it suitable for real-world lightweight FAS deployment.

III. METHODOLOGY

A. Overview

We propose *RepGhostFAS*, a lightweight and efficient face anti-spoofing (FAS) framework designed for deployment on IoT and edge devices. *RepGhostFAS* aims to effectively distinguish real and attack faces under diverse environmental and domain shift conditions while maintaining low computational cost. The proposed framework is utilized in three ways: (1) by employing the RepGhost backbone in RepGhostFAS Network for effective feature extraction and classification, (2) applying Ensemble-Based Feature Fusion to mitigate domain shift, and (3) adopting a Knowledge Distillation strategy to achieve a compact and efficient model.

B. RepGhostFAS Network

The RepGhost backbone [31] serves as the core feature extractor in the proposed *RepGhostFAS* framework. RepGhost extends the original GhostNet design [51] by introducing a

hardware-efficient RepGhost module through structural re-parameterization [57], enabling implicit feature reuse instead of relying on concatenation operations.

Different from conventional lightweight CNNs [47], [51], [58] that explicitly reuse features via concatenation across layers to generate more features cheaply, RepGhost performs feature reuse implicitly. Although concatenation is parameter- and FLOPs-free, its computational cost is non-negligible on hardware devices. As verified in [31], the concatenation operation is significantly less efficient than addition due to the complex memory copy process required during execution.

Formally, given an input image $I \in \mathbb{R}^{C_{in} \times H \times W}$ to be processed and reused, and $Y \in \mathbb{R}^{C_{out} \times H \times W}$ denote the output, where C_{in} and C_{out} are the number of input and output channels, and H and W represent spatial dimensions, let $\Phi_i, \forall i = 1, \dots, s-1$ denote the intermediate transformations (e.g., convolution or batch normalization) applied to I . Feature reuse via concatenation can be expressed as:

$$Y = Cat([I, \Phi_1(I), \dots, \Phi_{s-1}(I)]) \quad (1)$$

where $Cat(\cdot)$ represents the concatenation operation, which preserves existing feature maps and leaves the information processing to other layers. However, concatenation is not cost-free for feature reuse on hardware devices.

Structural re-parameterization has been demonstrated to be an efficient, cost-free technique for improving CNN performance in several studies [57], [59], [60]. As shown in [31], it can also serve as an implicit feature reuse, moving the fusion process from the feature space to weight space. During training, multiple linear operators are used to generate diverse feature maps, which are then fused into a single equivalent operator for fast inference. The feature reuse via structural re-parameterization can be formulated as:

$$Y = Add([I, \Phi_1(I), \dots, \Phi_{s-1}(I)]) = \Phi^*(I) \quad (2)$$

Unlike concatenation, the *Add* operation directly fuses features, and all linear functions $\Phi_i(\cdot)$ in structural re-parameterization are merged into $\Phi^*(I)$. The feature fusion is performed in the weight space, introducing no additional inference cost and yielding a more efficient architecture.

Finally, to adapt the network for the FAS task, the original classification layer of RepGhostNet [31] was replaced with a new fully connected layer for binary classification as illustrated in Fig. 2. Following [31], a width multiplier α is used to scale the network, denoted as *RepGhostFAS* $\alpha \times$ (default $1 \times$).

C. Ensemble-Based Feature Fusion

As highlighted in the previous sections, domain shift poses a significant challenge in FAS, since it is difficult to align features across different datasets due to discrepancies in data distribution. To address this issue, we propose three ensemble-based feature fusion strategies: concatenation, summation, and attention, as shown in Fig. 3. Each strategy aggregates features extracted from the frozen *RepGhostFAS Feature Extractor* for the datasets (source domain), and then the fused representation is fed into the classifier. These strategies are designed to learn complementary information

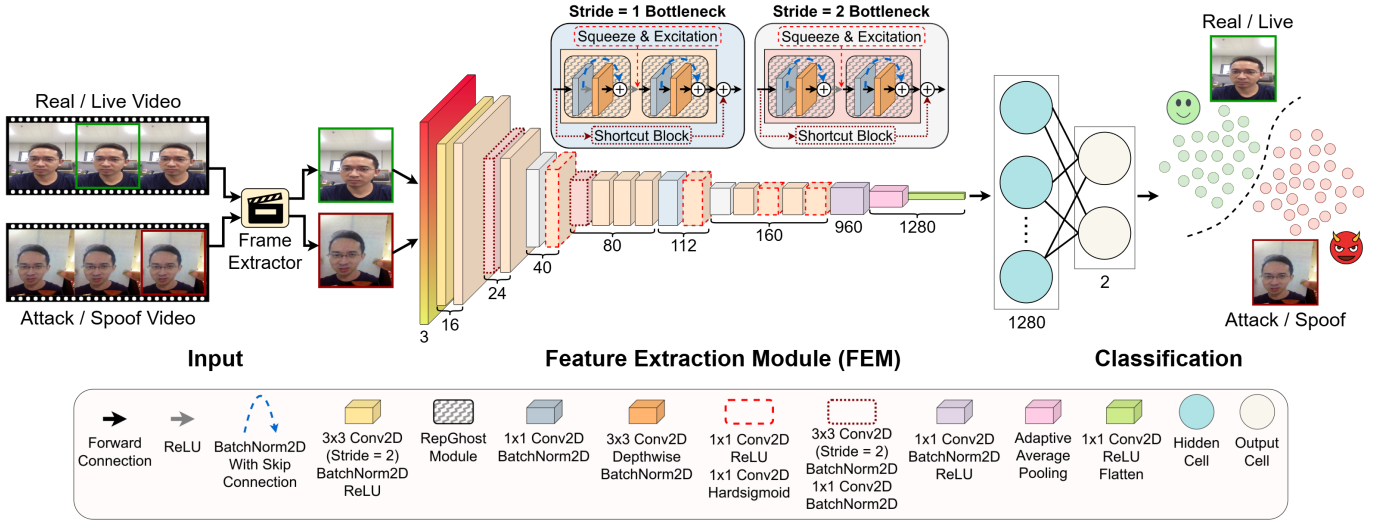


Fig. 2. Overview of the proposed *RepGhostFAS* architecture. The network extracts features using the RepGhost backbone and performs binary classification to distinguish real and attack faces. The blocks enclosed by dashed and dotted lines are inserted only when necessary.

from multiple domains and are described as follows:

Concatenation: The feature vectors $f_1, f_2, \dots, f_N$ produced by the frozen *RepGhostFAS Feature Extractor* are combined using channel-wise concatenation:

$$f_{cat} = Cat([f_1, f_2, \dots, f_N]) \quad (3)$$

where f_{cat} is the concatenated feature vector, which expands the feature space by preserving feature representations from all domains.

Summation: The same feature vectors $f_1, f_2, \dots, f_N$ are fused through element-wise addition:

$$f_{sum} = Add([f_1, f_2, \dots, f_N]) \quad (4)$$

This strategy produces a compact and efficient representation by emphasizing shared cross-domain features.

Attention: In the attention-based fusion, the feature vectors $f_1, f_2, \dots, f_N$ are first concatenated using Equation (3) and passed into the attention module Att_m , which learns feature-wise importance weights. The weighted features are then fused as:

$$f_{att} = \sum_{i=1}^N w_i f_i \quad (5)$$

where f_{att} allows the network to learn highly discriminative features while suppressing less informative ones.

D. Loss function

For the *RepGhostFAS* network, we employ the cross-entropy loss to measure the prediction error, defined as:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N [Y_i \log(\hat{Y}_i) + (1 - Y_i) \log(1 - \hat{Y}_i)] \quad (6)$$

where $Y_i \in \{0, 1\}$ represents the ground-truth labels (0 for real/live and 1 for attack/spoof) and $\hat{Y}_i$ denotes the model's estimated probability of the attack/spoof class.

E. Knowledge Distillation

To further balance performance and computational efficiency, we apply knowledge distillation [30] to transfer knowledge from the ensemble teacher model to a single *RepGhostFAS* student model. Knowledge distillation was first introduced by Hinton et al. [30]. It is a widely used model compression technique in which a compact student model learns to imitate the behavior of a larger and more expressive teacher model. The teacher model typically captures richer and more discriminative representations, and the goal is to transfer this knowledge to the student model, enabling it to achieve comparable performance with significantly fewer parameters.

Motivated by this idea, and based on our observation that the summation fusion strategy produces a compact and efficient representation, we adopt the *RepGhostFAS ensemble* model with summation fusion as the teacher, and train a single *RepGhostFAS* network as the student, as illustrated in Fig. 4. Let the teacher and student outputs be denoted by T_Y and S_Y , respectively. The objective is to minimize the discrepancy between the logit-based class probability distributions of the teacher and student through a KL-divergence loss:

$$\mathcal{L}_{KLD} = \mathcal{KL} \left[\text{softmax} \left(\frac{T_Y}{\tau} \right), \text{logsoftmax} \left(\frac{S_Y}{\tau} \right) \right] \cdot \tau^2 \quad (7)$$

where $\mathcal{KL}$ represents the Kullback-Leibler divergence, τ is the temperature parameter, and τ^2 is used as a multiplying factor as suggested in [30].

In addition, we compute the mean squared error (MSE) loss between the intermediate feature representations of the teacher and student, which encourages the student to align internal feature structures with those of the teacher:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N (f_T - f_S)^2 \quad (8)$$

Finally, the student is supervised by the cross-entropy (CE)

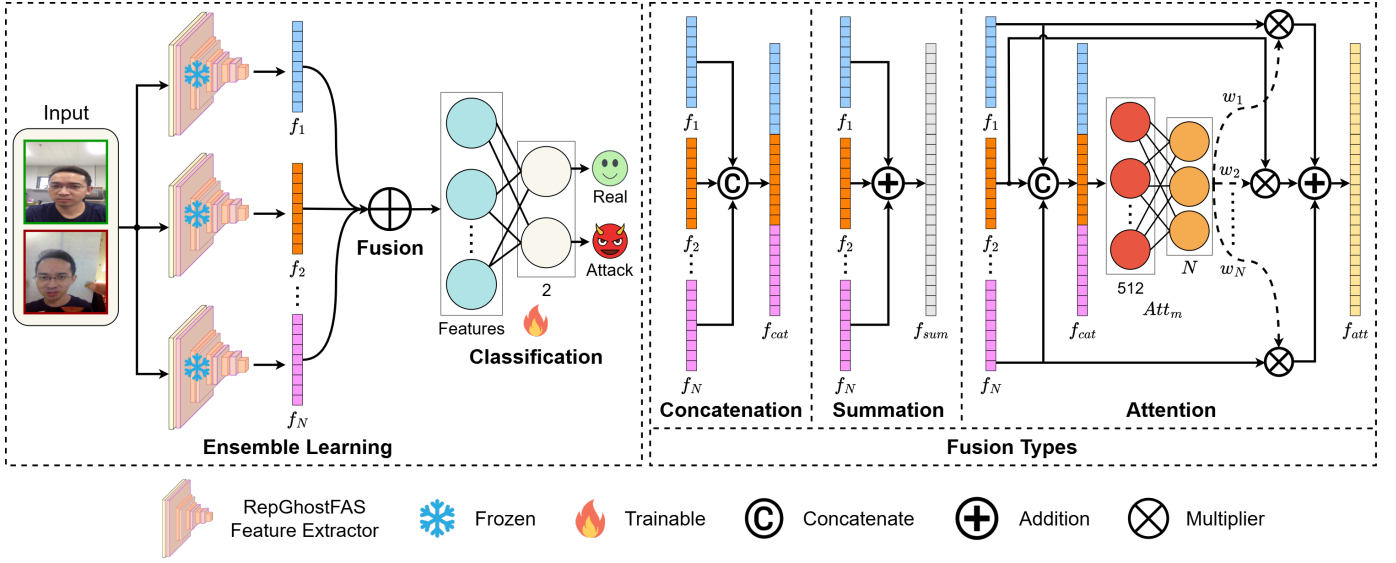


Fig. 3. Ensemble-based feature fusion in the proposed *RepGhostFAS*. Features extracted from N domain-specific branches are fused using concatenation, summation, or attention before final binary classification.

loss with respect to the ground-truth labels Y_i :

$$\mathcal{L}'_{CE} = -\frac{1}{N} \sum_{i=1}^N [Y_i \log(S_Y) + (1 - Y_i) \log(1 - S_Y)] \quad (9)$$

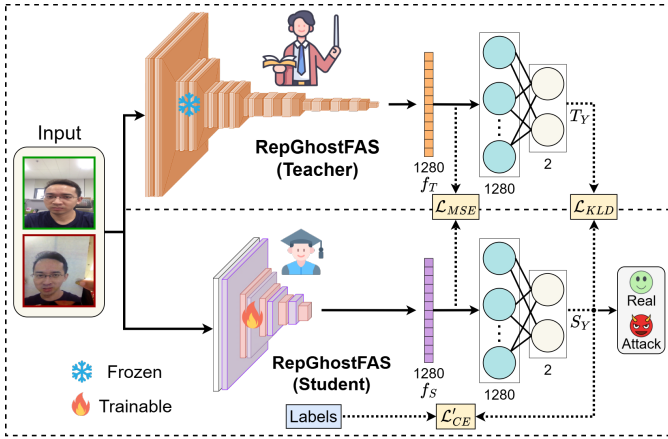


Fig. 4. Knowledge distillation process for *RepGhostFAS*. The teacher ensemble model (summation fusion) provides logits and feature guidance to train a compact student model using KLD, MSE, and CE losses.

During the training of *student model*, the teacher network remains frozen, and only the student parameters are optimized. The overall training objective combines all supervision signals through a knowledge distillation loss:

$$\mathcal{L}_{KD} = \lambda_1 \mathcal{L}_{KLD} + \lambda_2 \mathcal{L}_{MSE} + \lambda_3 \mathcal{L}'_{CE} \quad (10)$$

where λ_1 , λ_2 , and λ_3 are hyperparameters that balance the contribution of divergence loss, feature alignment loss, and classification loss.

IV. EXPERIMENTAL SETUP

A. Datasets

The proposed method is evaluated on eight publicly available benchmark face anti-spoofing datasets: Replay-Attack

[32], Replay-Mobile [61], OULU-NPU [62], ROSE-Youtu [20], SiW-Mv2 [22], SynthASpoof [63], SOTERIA [64], and UniAttackData [65]. These datasets cover a broad range of subjects, environmental conditions, camera devices, and presentation attack types. In particular, OULU-NPU and ROSE-Youtu are considered particularly challenging due to their higher diversity and wider variety of spoofing scenarios. A brief summary of all datasets is provided in Table I, and Fig. 5 shows some real and attack samples from each dataset.

TABLE I
FACE ANTI-SPOOFING DATASETS DESCRIPTION.

Datasets	Attack Types	Subjects		Samples	
		Train	Test	Real	Attack
Replay-Attack [32]	Print (Photo), Replay (Video)	30	20	200	1,000
Replay-Mobile [61]	Print (Photo), Replay (Display)	23	16	380	624
OULU-NPU [62]	Print (Photo), Replay (Phone)	35	20	990	3,960
ROSE-Youtu [20]	Print (Photo), Replay (Display, Laptop), Mask (Paper, Crop)	10	10	897	2,600
SiW-Mv2 [22]	Print (Photo), Replay (Video), 3D Mask, Obfuscation	874	219	785	915
SynthASpoof [63]	Print (Photo), Replay (Photo)	25,000		-	78,000
SOTERIA [64]	Print (Photo), Replay (Video, TV, Projector)	70		-	24,265
UniAttackData [65]	Print (Photo), Replay (Video), Digital Forgery, Adversarial Attack	1,800		-	27,906

Replay-Attack: This dataset contains 1,200 videos from 50 subjects, each lasting 9 – 15 seconds and recorded at 25 FPS with a resolution of 320×240 under two lighting conditions (controlled and adverse). The real videos were recorded using a MacBook laptop, while the attack samples were created with a Canon PowerShot SX150 IS camera and an iPhone 3GS. Attack is divided into fixed-support and hand-held subsets. The training and validation sets include data from 15 subjects, and the test set comprises data from 20 subjects.

Replay-Mobile: Consists of 1,004 videos from 39 subjects, recorded with the front cameras of mobile devices at 25 FPS and 720×1280 resolution. Each clip lasts about 10 seconds.

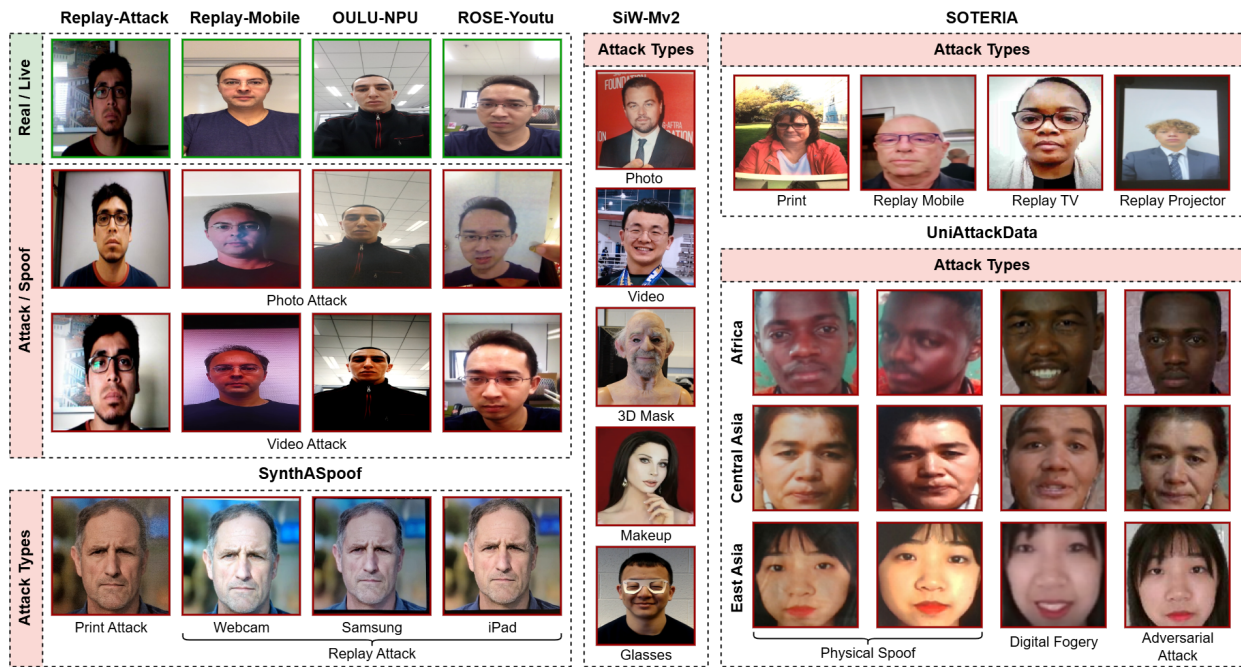


Fig. 5. Visualization of samples from the Replay-Attack [32], Replay-Mobile [61], OULU-NPU [62], ROSE-Youtu [20], SynthASpoof [63], SiW-Mv2 [22], SOTERIA [64], and UniAttackData [65]. Each dataset exhibits unique variations in attack types, illumination, camera devices, and background conditions.

Photo attacks were created using a Nikon Coolpix P520 (18 MP) camera, and video attacks were recorded with an LG G4 smartphone (1080p), both displayed on a Philips 227ELH monitor (1920 × 1080). The dataset is split into training (11 subjects), validation (12 subjects), and testing (16 subjects) sets.

OULU-NPU: Includes 4,950 videos from 55 subjects recorded using six mobile phones (HTC Desire Eye, ASUS Zenfone Selfie, Oppo N3, Meizu MX5, Sony Xperia C5 Ultra Dual, and Samsung Galaxy S6 edge) across three sessions with varying background scenes and lighting conditions. Two printers and two display devices were used to create presentation attacks. The dataset is divided into training and testing sets, each containing data from 20 subjects, while 15 subjects are used for validation.

ROSE-Youtu: The ROSE-Youtu is a large-scale industry collected dataset comprising 3,497 videos from 20 subjects, captured using five devices (Hasee, Huawei, iPhone 5s, ZTE, and iPad) with resolutions from 640 × 480 to 1280 × 720. In addition to photo and video replay attacks, this dataset also includes paper mask attacks under five different lighting conditions. Photo attacks were produced using a printer, replay attacks with Lenovo LCD and Mac screens, and masking attacks with cropped and full-face masks. Training uses 8 subjects, validation 2 subjects, and testing 10 subjects.

SynthASpoof: SynthASpoof is a synthetically generated dataset designed to reflect realistic spoofing conditions. It contains 78,800 high-quality attack images from 25,000 subjects covering two attack types: print and replay. For

print attacks, 3,800 synthetic identities were captured using a Samsung Galaxy Tablet S6. For replay attacks, one set was displayed on a Dell UltraSharp 24 monitor and recorded with a Creative Labs webcam at 720 × 480, while two additional sets were displayed on a MacBook Air 2020 and captured using a Samsung Galaxy A71 and an iPad Pro 10.5 at 1920 × 1080.

SiW-Mv2: Contains 1,700 videos from 1,093 subjects. It includes 14 attack types, ranging from print and replay attacks to 3D masks and obfuscation attacks (e.g., makeup, covered eyes, glasses). These attacks introduce diverse spoof patterns designed to impersonate a user's identity.

SOTERIA: Focuses on the mobile face verification scenario in face anti-spoofing. It consists of 24,265 attack videos from 70 subjects captured using both the front and back cameras of nine mobile phones (Apple iPhone 6s, 7 Plus, 12, 12 Mini; Google Pixel 3; Samsung Galaxy S8, S9; and Xiaomi Redmi 6 Pro, 9A). The videos are recorded under different lighting conditions, including normal, low, and lateral illumination.

UniAttackData: Provides a complete set of attack types for each individual. It includes three ethnic groups (African, East Asian, and Central Asian). The dataset comprises 27,906 videos from 1,800 subjects, including 2 physical attacks (print and replay) and 12 digital attacks. The digital attacks are generated using digital editing techniques and adversarial algorithms. Although this demographic diversity broadens the evaluation coverage, our experiments follow the standard aggregate and attack-wise evaluation protocol and do not report performance separately for each ethnic or gender group.

B. Implementation Details

For all experiments, input images were resized to 256×256 pixels as the standard configuration. During training, a random frame $I \in \mathbb{R}^{C_{in} \times H \times W}$ was extracted from each video to introduce greater variability. All models were initialized with ImageNet-pretrained weights [66] and trained on the respective datasets for 50 epochs. Implementation is performed using Pytorch framework, Adam optimizer with 0.001 learning rate, and 64 batch size. To prevent overfitting, early stopping (patience = 5) was applied, terminating training when the validation loss stopped improving. Temperature parameter τ in Equation (7) was set to 3.0, and the hyperparameters λ_1 , λ_2 , and λ_3 in Equation (10) were empirically set to 0.2, 0.3, and 0.5, respectively. During testing, a fixed frame at time step ($t = 5$) is selected to ensure consistent and fair evaluation. Code is available at <https://github.com/tahahasanmasood/RepGhostFAS>. A detailed analysis of model complexity, including parameters, FLOPs, inference latency, and memory consumption across different hardware platforms, is provided in Section V-H.

C. Evaluation Metrics

We employ standard evaluation metrics commonly used in previous studies. For the Replay-Attack, Replay-Mobile, and ROSE-Youtu datasets, we adopt the Half Total Error Rate (HTER) and Equal Error Rate (EER), defined as follows:

$$HTER = \frac{FAR + FRR}{2}, \quad EER = |FAR - FRR|$$

where FAR and FRR denote the False Acceptance Rate and False Rejection Rate, respectively.

For the OULU-NPU dataset, we follow the recently standardized ISO/IEC 30107-3 metrics [67], including the Attack Presentation Classification Error Rate (APCER), Bonafide Presentation Classification Error Rate (BPCER), and Average Classification Error Rate (ACER), given as follows:

$$APCER = \frac{FN}{TP + FN}, \quad BPCER = \frac{FP}{FP + TN},$$

$$ACER = \frac{APCER + BPCER}{2}$$

Here, TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives.

V. EXPERIMENTAL RESULTS

A. Intra-Dataset Results

The proposed *RepGhostFAS* is evaluated under an intra-dataset setting across four benchmarks, and the results in Table II highlight its strong and consistent performance. On Replay-Attack and Replay-Mobile, the model achieves perfect classification with 100% accuracy and 0% HTER/EER, reflecting its ability to capture spoofing cues with complete reliability. Performance remains high on the more challenging OULU-NPU and ROSE-Youtu datasets, where *RepGhostFAS* attains 99.66% and 99.25% accuracy, respectively, alongside low HTER and EER values. These results underscore the model's robustness in handling diverse environments and complex attack variations.

TABLE II
INTRA-DATASET TESTING RESULTS ON FOUR DATASETS.

Metric	Replay-Attack	Replay-Mobile	OULU-NPU	ROSE-Youtu
Test Accuracy (%)	100	100	99.66	99.25
HTER (%)	0.0	0.0	0.31	1.30
EER (%)	0.0	0.0	0.33	0.74

B. Comparison with State-of-the-Art Methods

We compare the proposed *RepGhostFAS* models ($1 \times$ and $2 \times$) with recent state-of-the-art (SOTA) methods across all benchmark datasets. *RepGhostFAS* $1 \times$ is adopted as the default configuration, offering a strong balance between accuracy and computational cost, while *RepGhostFAS* $2 \times$ is a moderately larger variant with increased capacity. For fair comparison, we follow standard evaluation metrics, including HTER, EER, APCER, BPCER, and ACER, where lower values indicate better performance.

Table III reports the results on the Replay-Attack and Replay-Mobile. Both *RepGhostFAS* variants achieve 0.0% HTER on these benchmarks, eliminating all classification errors and outperforming most existing SOTA methods. Table IV presents a detailed comparison on OULU-NPU: *RepGhostFAS* $1 \times$ attains 0.0% APCER, 0.83% BPCER, and 0.41% ACER, providing performance comparable to AdvSpoofGuard [68] at substantially lower computational cost. The larger *RepGhostFAS* $2 \times$ further improves the results to 0.13% APCER, 0.55% BPCER, and an ACER of 0.34%. In addition, Table V compares the protocol-wise performance on OULU-NPU, where some methods perform slightly better on individual protocols. *RepGhostFAS* $1 \times$ maintains consistent ACER across both protocols, while *RepGhostFAS* $2 \times$ further reduces ACER to 1.4%, marking a clear gain over previously reported results. On the ROSE-Youtu (Table VI), which includes diverse spoof types and challenging environmental conditions, *RepGhostFAS* $1 \times$ remains competitive with an HTER/EER of 1.30%/0.74%, while *RepGhostFAS* $2 \times$ achieves the best overall performance, reaching the lowest HTER/EER of 0.67%/0.45% among all compared methods. Moreover, consistent trends are observed on the SiW-Mv2 protocol I (Table VII). Here again, *RepGhostFAS* $1 \times$ maintains competitive performance, while *RepGhostFAS* $2 \times$ obtains HTER/EER of 3.38%/3.31%, outperforming all other methods.

C. Latent Space Visualization

Fig. 6 presents visualization of the latent space using t-SNE [107] across four datasets. In the image domain (top row), the embeddings of real and attack samples exhibit substantial overlap, indicating weak discriminability and showing that it is difficult to separate these classes when relying directly on the raw image data. In contrast, the feature space learned by the proposed *RepGhostFAS* (bottom row) shows a much clearer separation between real and attack samples. The improved clustering structure demonstrates that the model effectively learns discriminative representations, resulting in a more distinct decision boundary and enhanced spoof detection performance.

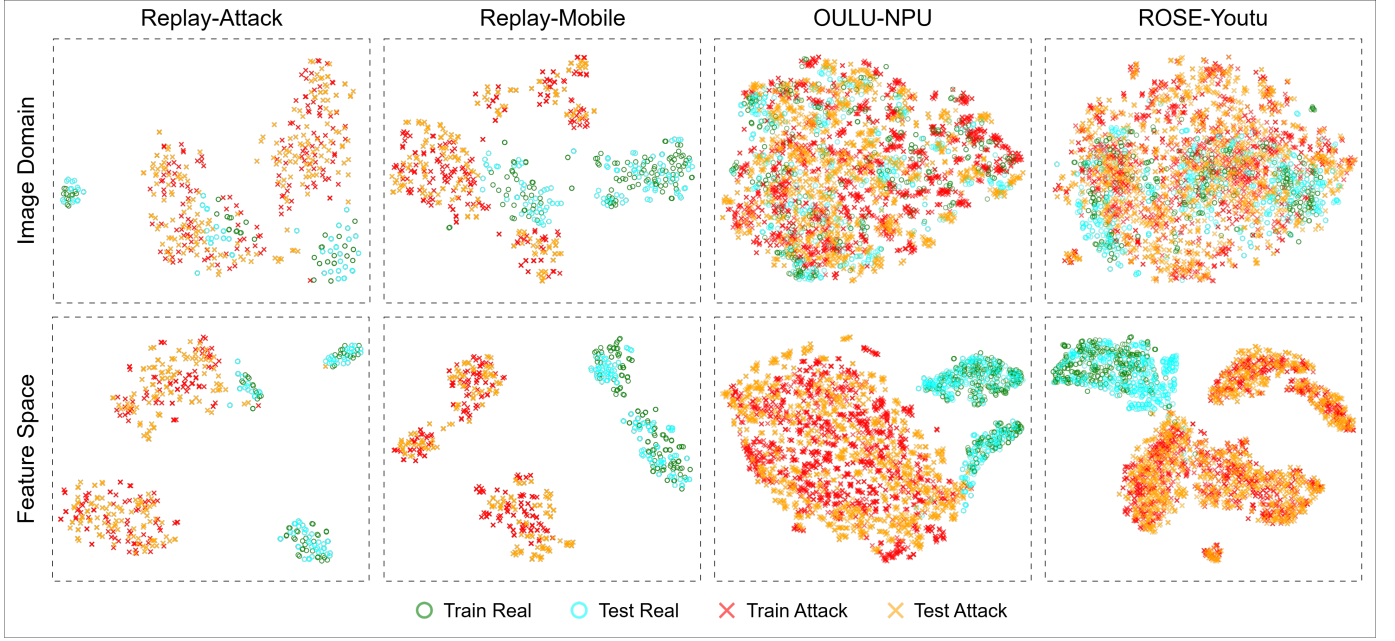


Fig. 6. Latent space visualization using t-SNE [107] comparing data distributions in the image and feature domains. The top row shows the original dataset samples in the image domain, while the bottom row illustrates the feature representations learned by the proposed *RepGhostFAS* model.

TABLE III

PERFORMANCE COMPARISON ON REPLAY-ATTACK AND REPLAY-MOBILE TEST DATASETS.

Method	Year	HTER (%)	
		Replay-Attack	Replay-Mobile
GoogLeNet + GMM [69]	2021	3.76	13.56
VGG16 + GMM [69]	2021	1.46	17.21
WA (PSO + PS) [70]	2021	0.0	5.85
WA (GA + MMS + PS) [71]	2022	0.0	5.35
MobileNet + Image Diffusion [72]	2023	0.09	1.14
Localised MKL [73]	2023	-	5.6
ResNet50V2 [74]	2023	0.03	0.0
Lightweight 3D-DNN [75]	2024	0.0	0.45
EfficientNet-B0 [76]	2024	36.88	4.62
3D ConvNet [77]	2024	11.7	8.7
Spatio-Temporal [78]	2025	0.0	0.0
LwFLNet [79]	2025	2.12	-
Deformable Convolution [80]	2025	0.0	0.0
AdvSpooGuard [68]	2025	0.0	0.0
KD + Depth [81]	2025	0.0	0.0
RepGhostFAS 1× (Ours)	2026	0.0	0.0
RepGhostFAS 2× (Ours)	2026	0.0	0.0

D. Class Activation Map Visualization

To better understand the regions influencing the classification decisions, we visualize the class activation maps using Grad-CAM [108]. Grad-CAM leverages the gradient information flowing into the last convolutional layer to highlight which parts of the input image contribute most to the model's decision. Fig. 7 illustrates the activated regions for real and attack samples across four datasets. The proposed *RepGhostFAS* focuses on meaningful facial regions, which contain the most informative cues. This indicates that the model has learned to extract discriminative features that contribute to reliable spoof

TABLE IV

PERFORMANCE COMPARISON ON OULU-NPU (COMPLETE) TEST DATASET.

Method	Year	APCER (%)	BPCER (%)	ACER (%)
Texture (VAR) [82]	2021	14.5	15	14.8
Fake-Net (VAR) [83]	2021	5.4	6.9	6.2
ED-LBP (VAR) [84]	2021	11.3	8.4	9.9
OFT (VAR) [85]	2022	5.7	2.7	4.2
ELA [86]	2023	-	-	4.86
UCDCN [87]	2024	2.6	1.01	1.82
3DLCN [88]	2024	1.5	0.5	1.0
Spatio-Temporal [78]	2025	0.13	1.11	0.62
KD + Depth [81]	2025	0.28	0.83	0.56
AdvSpooGuard [68] w/o AT	2025	0.6	0.2	0.4
RepGhostFAS 1× (Ours)	2026	0.0	0.83	0.41
RepGhostFAS 2× (Ours)	2026	0.13	0.55	0.34

detection.

E. Domain Shift Analysis

In order to analyze domain shift, we consider three datasets, namely Replay-Attack (A), Replay-Mobile (M), and ROSE-Youtu (Y). The model is trained on the combined datasets and evaluated on each individual dataset using the AMY→A, AMY→M, and AMY→Y protocol. The corresponding results are summarized in Table VIII.

The *RepGhostFAS (Baseline)* model exhibits a noticeable drop in performance when tested on individual dataset, confirming the severity of domain shift. To address this issue, we introduce three ensemble-based feature fusion strategies namely concatenation, summation, and attention within the *RepGhostFAS Ensemble* model. Concatenation increases feature dimensionality and parameters, leading to redundant

TABLE V
PERFORMANCE COMPARISON ON OULU-NPU (PROTOCOL I AND II) TEST DATASET.

Method	Year	Protocol I			Protocol II			Avg. ACER (%)
		APCER (%)	BPCER (%)	ACER (%)	APCER (%)	BPCER (%)	ACER (%)	
Bi-FAS [89]	2021	2.9	3.3	3.1	2.3	1.1	1.7	2.4
Camera Invariant [90]	2021	3.8	2.9	3.4	3.6	1.2	2.4	2.9
MIQF + SVM [91]	2022	6.9	1.5	4.2	7.8	1.4	4.6	4.4
PFTD-MHSA [92]	2022	2.4	14.2	8.3	18.3	11.3	14.8	11.5
TSS with ResNet [93]	2022	0.6	10.3	5.5	2	2.1	2.1	3.8
MSCI-DSCNN [94]	2023	0.3	2.9	1.6	3.1	7.2	5.2	3.4
CSN-IR [95]	2023	3.7	6.7	5.2	0.0	3.6	1.8	3.5
M3D [95]	2023	0.0	1.7	0.8	9.2	0.0	4.6	2.7
AlexNet + PNFG [14]	2024	0.0	63.2	31.6	5.0	10.4	7.7	19.6
AdvSpoofGuard [68] w/o AT	2025	1.6	5	3.3	5.2	0	2.6	2.9
C&T-LSTM [96]	2025	-	-	9.7	-	-	7.6	8.6
SR&CD [97]	2025	3.7	1.4	2.5	3.4	1.3	2.4	2.4
RepGhostFAS 1× (Ours)	2026	0.0	3.3	1.6	3.8	1.1	2.5	2.0
RepGhostFAS 2× (Ours)	2026	0.4	1.6	1.0	3.3	0.5	1.9	1.4

TABLE VI
PERFORMANCE COMPARISON ON ROSE-YOUTU TEST DATASET.

Method	Year	HTER/EER (%)
FASNet [98]	2021	8.57/-
RCTR-All Spaces [99]	2021	-/10.7
ResNet50 + GMM [69]	2021	14.69/-
AlexNet [100],	2022	-/8.0
WA (GA + MMS + PS) [71]	2022	5.12/-
OCC Fusion [101]	2022	6.34/-
MobileNet+Image Diffusion [72]	2023	4.92/4.95
ViViT [102]	2023	13.28/2.46
ResNet50V2 [74]	2023	2.53/2.64
Hyp-OC [103]	2024	34.07/-
3D ConvNet [77]	2024	21.3/-
EfficientNet-B0 [76]	2024	9.54/-
CA-FAS [104]	2024	14.88/-
Spatio-Temporal ($\tau = 5$) [78]	2025	1.54/0.87
AdvSpoofGuard [68]	2025	1.97/1.08
Deformable Convolution [80]	2025	1.26/0.80
KD + Depth [81]	2025	1.02/1.15
RepGhostFAS 1× (Ours)	2026	1.30/0.74
RepGhostFAS 2× (Ours)	2026	0.67/0.45

TABLE VII
PERFORMANCE COMPARISON ON SiW-MV2 (PROTOCOL I) TEST DATASET.

Method	Year	HTER/EER (%)
VGG16 [105]	2014	10.54/10.45
InceptionV3 [106]	2016	5.77/5.78
MobileNetV3 [50]	2019	5.93/5.87
Spatio-Temporal ($\tau = 1$) [78]	2025	6.11/6.63
KD + Depth [81]	2025	5.82/6.08
RepGhostFAS 1× (Ours)	2026	5.32/5.82
RepGhostFAS 2× (Ours)	2026	3.38/3.31

representations and unnecessary complexity. Attention fusion provides adaptive feature weighting but limits the preservation of complementary domain-specific features. In contrast,

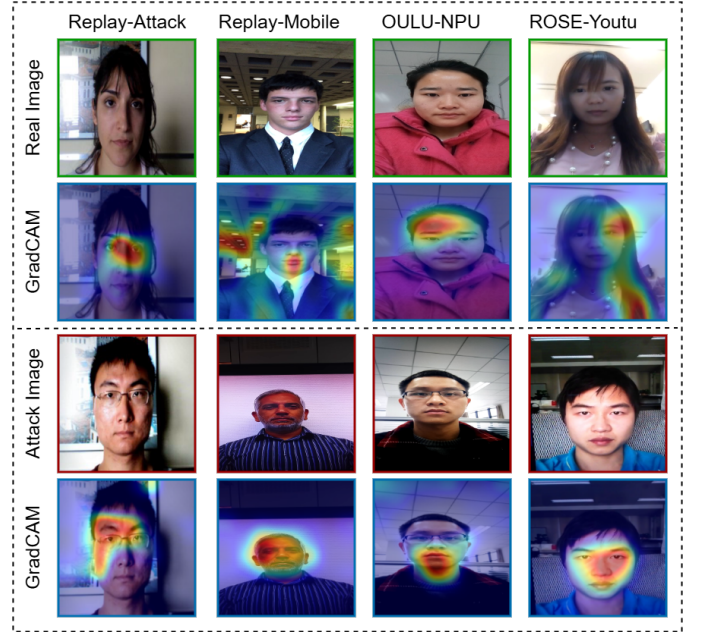


Fig. 7. Visualization of attention map using GradCAM [108] from the proposed *RepGhostFAS*. The model focuses more on meaningful facial regions for both real and attack faces, indicating its effective discrimination capability.

summation fusion effectively combines complementary representations while maintaining the original feature dimension, enabling compact and efficient knowledge transfer. All three fusion strategies lead to substantial performance improvements over the baseline. Among them, the summation-based fusion provides the most significant gains, reducing HTER by 80.08% and EER by 66.07%. This shows that summation fusion is particularly effective in aggregating complementary cross-domain representations.

Based on these findings, we select the *RepGhostFAS Ensemble (Summation)* model as the teacher for the knowledge distillation. The goal is to transfer its discriminative capability to a single *RepGhostFAS (Student)* model with much lower computational cost. As shown in Table VIII, the distilled *student model* achieves slightly higher HTER and EER val-

TABLE VIII
DOMAIN SHIFT ANALYSIS (TRAIN $\rightarrow$ TEST). LOWER HTER (%), EER (%) VALUES INDICATE BETTER PERFORMANCE.

Model Dataset	RepGhostFAS (Baseline)		RepGhostFAS Ensemble (Concatenation)		RepGhostFAS Ensemble (Summation) §		RepGhostFAS Ensemble (Attention)		RepGhostFAS (Student)	
	HTER (%)	EER (%)	HTER (%)	EER (%)	HTER (%)	EER (%)	HTER (%)	EER (%)	HTER (%)	EER (%)
AMY $\rightarrow$ A	3.12	1.04	0.75	1.25	0.0	0.0	0.0	0.0	0.0	0.0
AMY $\rightarrow$ M	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AMY $\rightarrow$ Y	3.98	2.34	1.83	1.49	1.42	1.14	1.83	1.31	2.31	1.48
Avg.	2.36	1.12	0.86	0.91	0.47	0.38	0.61	0.43	0.77	0.49

AMY: Combined Datasets, § RepGhostFAS (Teacher).

ues than the teacher, which is expected due to its smaller capacity. However, it still outperforms the baseline model by a significant improvement while using only one-third of the parameters and FLOPs. This confirms that the knowledge distillation provides an effective balance between performance and efficiency while preserving generalization ability under domain shift conditions.

F. Ablation Study

To verify the effectiveness of the proposed *RepGhostFAS*, we conduct an ablation study on the ROSE-Youtu dataset. This dataset is selected because it is the most challenging benchmark due to its diversity and wide range of attack scenarios.

Impact of Network Scaling: We investigate how scaling the network width affects the performance and computational cost of *RepGhostFAS*. The results in Table IX show three configurations: *RepGhostFAS* 0.5 $\times$, 1 $\times$, and 2 $\times$. The smallest configuration, *RepGhostFAS* 0.5 $\times$, achieves the lowest parameters and floating-point operations per second (FLOPs), reflecting its computational efficiency. However, this reduction in model capacity leads to decline in performance. In contrast, *RepGhostFAS* 2 $\times$ improves accuracy, HTER, and EER, but the performance gain is modest relative to the substantial increase in model size and computational cost. Among the three variants, *RepGhostFAS* 1 $\times$ offers the favorable balance between accuracy and efficiency. It maintains high detection performance while keeping both parameters and FLOPs count at a moderate level.

TABLE IX
PERFORMANCE COMPARISON OF REP-GHOSTFAS WITH DIFFERENT WIDTH MULTIPLIERS.

Model Metric	RepGhostFAS 0.5 $\times$	RepGhostFAS 1 $\times$	RepGhostFAS 2 $\times$
Image Size	256 $\times$ 256	256 $\times$ 256	256 $\times$ 256
Parameters (M)	1.03	2.79	8.57
FLOPs (G)	0.12	0.38	1.37
Test Accuracy (%)	98.74	99.25	99.54
HTER (%)	2.30	1.30	0.67
EER (%)	1.25	0.74	0.45

Impact of Image Size: We evaluate how different input image sizes influence the performance and computational cost of *RepGhostFAS* 1 $\times$. As shown in Table X, lower resolutions

such as 64 $\times$ 64 and 128 $\times$ 128 require very few FLOPs but lead to noticeably degraded performance, indicating that important facial details are lost at smaller scales. In contrast, the higher resolution input of 512 $\times$ 512 achieves the highest accuracy and lowest error rates but incurs a substantial increase in FLOPs. The input size of 256 $\times$ 256 provides the best trade-off with strong detection performance while keeping the FLOPs within acceptable limit.

TABLE X
PERFORMANCE COMPARISON OF REP-GHOSTFAS WITH DIFFERENT IMAGE INPUT SIZES.

Image Size Metric	64 $\times$ 64	128 $\times$ 128	256 $\times$ 256	512 $\times$ 512
Parameters (M)	2.79	2.79	2.79	2.79
FLOPs (G)	0.03	0.10	0.38	1.52
Test Accuracy (%)	95.14	97.59	99.25	99.48
HTER (%)	8.44	4.38	1.30	0.63
EER (%)	4.85	2.40	0.74	0.51

Selection of RepGhostFAS (Teacher): To further validate the selection of the RepGhostFAS Ensemble (Summation) model as the *RepGhostFAS (Teacher)* for knowledge distillation (refer to Subsection V-E), we extend the experiment by using concatenation-based and attention-based fusion strategies as teacher models. Both teacher models transfer their knowledge to the *RepGhostFAS (Student)*, and their performance is evaluated in Table XI under the same experimental settings. As observed, both concatenation and attention fusion underperform compared with summation fusion. These results justify the selection of the summation-based *RepGhostFAS (Teacher)*, which achieves superior performance and provides more effective feature aggregation for knowledge transfer.

TABLE XI
COMPARISON OF ENSEMBLE-BASED FEATURE FUSION STRATEGIES FOR SELECTING THE REP-GHOSTFAS (TEACHER) MODEL.

RepGhostFAS (Teacher) Ensemble Fusion	RepGhostFAS (Student)	
	HTER (%)	EER (%)
Concatenation	3.07	3.48
Attention	2.92	3.37
Summation	2.31	1.48

Failure Case Analysis: To analyze the performance gap between the *RepGhostFAS (Teacher)* and *RepGhostFAS (Student)* models, we examine misclassified samples as summarized in Table XII. The *student model* produces fewer attack errors

(8 vs. 11) but significantly more real errors (18 vs. 9), indicating a tendency toward more conservative spoof detection. Most failures are associated with variations in recording devices, environmental conditions, subject appearance, and attack types. For attack samples, misclassifications are mainly related to high-quality photo and replay attacks, suggesting that subtle spoofing cues remain challenging to capture. For real samples, the *student model* is more sensitive to device-specific variations (e.g., ZTE, iPad, Hasee, Huawei), as well as conditions such as non-speaking scenarios and the absence of glasses. These results clarify that the performance gap is primarily due to reduced representational capacity, which limits robustness to fine-grained variations.

TABLE XII
FAILURE CASE ANALYSIS OF REP_{GH}STFAS (TEACHER & STUDENT),
SHOWING MISCLASSIFICATION COUNT AND KEY FACTORS.

Model	Real Errors	Real Failure Cases	Attack Errors	Attack Failure Cases
Rep _{GH} STFAS (Teacher)	9	Device (ZTE, iPad), Environment, Without Glasses, Non-Speaking	11	Replay (Mac/Lenovo), Still Photo
Rep _{GH} STFAS (Student)	18	Device (ZTE, iPad Hasee, Huawei), Environment, Without Glasses, Non-Speaking	8	Replay (Mac), Still Photo

Hyperparameter Sensitivity Analysis: The loss balancing hyperparameters in Equation (10) were empirically set as $\lambda_1 = 0.2$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.5$, assigning higher importance to the ground-truth classification loss ($\mathcal{L}'_{CE}$) while retaining complementary supervision from *teacher predictions* and *feature representations*. This balance enables the lightweight Rep_{GH}STFAS (*Student*) model to preserve discriminative capability (real vs. attack) while benefiting from the knowledge transferred by the Rep_{GH}STFAS (*Teacher*) model. We investigate the sensitivity of these hyperparameters by varying each individual loss weight by ($\pm$) while keeping the remaining loss terms unchanged. The results in Table XIII demonstrate that the selected weights provide a favorable trade-off between knowledge transfer and classification supervision. Increasing or decreasing the contribution of individual loss terms leads to performance degradation, highlighting the importance of appropriate loss balancing.

TABLE XIII
EFFECT OF HYPERPARAMETERS SELECTION ON REP_{GH}STFAS
(STUDENT) PERFORMANCE.

Hyperparameters			Rep _{GH} STFAS (Student)	
λ_1	λ_2	λ_3	HTER (%)	EER (%)
0.3	0.3	0.5	3.80	3.37
0.1	0.3	0.5	2.60	3.20
0.2	0.4	0.5	3.51	3.60
0.2	0.2	0.5	2.43	2.97
0.2	0.3	0.6	2.68	2.69
0.2	0.3	0.4	3.30	2.85
0.2	0.3	0.5	2.31	1.48

Rep_{GH}STFAS (Teacher): Rep_{GH}STFAS Ensemble (Summation).

Knowledge Distillation Loss Analysis: To investigate the contribution of each component in the knowledge distillation loss ($\mathcal{L}_{KD}$), we progressively incorporate individual loss terms, including classification loss ($\mathcal{L}'_{CE}$), logit-level KL-divergence loss ($\mathcal{L}_{KLD}$), and feature alignment loss ($\mathcal{L}_{MSE}$) under identical Rep_{GH}STFAS (*Teacher*) and Rep_{GH}STFAS (*Student*) settings. The results presented in Table XIV show that using only the classification loss provides performance with an HTER of 2.89% and an EER of 2.45%. Incorporating feature-level knowledge through $\mathcal{L}_{MSE}$ improves performance by transferring intermediate representations from the teacher model. Similarly, adding logit-level knowledge through $\mathcal{L}_{KLD}$ further enhances the *student model* performance. Finally, combining all loss components achieves the best performance with an HTER of 2.31% and an EER of 1.48%, demonstrating that output-level and feature-level knowledge provide complementary benefits for effective knowledge transfer.

TABLE XIV
EFFECT OF KNOWLEDGE DISTILLATION LOSS COMPONENTS ON
REP_{GH}STFAS (STUDENT) PERFORMANCE.

$\mathcal{L}_{KD}$			Rep _{GH} STFAS (Student)	
$\mathcal{L}'_{CE}$	$\mathcal{L}_{KLD}$	$\mathcal{L}_{MSE}$	HTER (%)	EER (%)
✓	✗	✗	2.89	2.45
✓	✗	✓	2.75	1.77
✓	✓	✗	2.35	1.71
✓	✓	✓	2.31	1.48

Rep_{GH}STFAS (Teacher): Rep_{GH}STFAS Ensemble (Summation).

G. Cross-Dataset Results

We evaluate the generalization capability of the proposed Rep_{GH}STFAS (*Student*) by comparing it against several lightweight models on unseen datasets (SynthASpoof [63], SOTERIA [64], UniAttackData [65]). Table XV presents the cross-dataset evaluation results. As expected, all other models trained on the combined A, M, and Y datasets exhibit performance degradation when tested on unseen datasets due to the significant domain shift. Despite this, Rep_{GH}STFAS (*Student*) achieves the lowest EER (%) across all attack types, outperforming other models by a clear margin. These results highlight that Rep_{GH}STFAS (*Student*) provides strong robustness against unseen spoofing attacks and maintains high reliability under challenging cross-domain generalization.

H. Computational Complexity

We compare the computation complexity of Rep_{GH}STFAS with several lightweight models. The evaluation considers multiple factors including model size (parameters), FLOPs, latency, and memory usage which are crucial indicators for deployment on resource-constrained devices. Fig. 8 presents the relationship between HTER versus number of parameters (left), and HTER versus number of FLOPs (right). From the plots, we can see that ShuffleNetV2 [47] and Once-for-All [48] exhibit high HTER values, showing that extreme compression severely compromises performance. ViT models (MobileViTV2, FastViT [53], SHViT-S1 [54], SwiftFormer

TABLE XV
CROSS-DATASET TESTING RESULTS. LOWER EER (%) VALUES INDICATE BETTER RESULTS.

Test Train (AMY)	Attack Types															Avg.		
	SynthASpoof				SOTERIA				UniAttackData									
	Print	Replay Attack (iPAD)	Replay Attack (Samsung)	Replay Attack (Webcam)	Print	Replay Mobile	Replay TV	Projector	Physical Spoof			Digital Forgery			Adversarial Attack			
									Africa	Central Asia	East Asia	Africa	Central Asia	East Asia	Africa		Central Asia	East Asia
ShuffleNetV2 [47]	0.21	0.78	0.61	0.43	2.87	0.63	4.42	23.47	0.86	0.0	1.10	1.64	0.22	2.30	1.05	0.0	4.52	2.65
Once-for-All [48]	1.97	2.47	2.78	0.03	3.51	0.16	8.19	2.53	0.70	0.0	1.50	7.77	5.52	14.09	0.0	0.0	0.02	3.01
MobileViTV2 [52]	1.61	0.27	17.11	5.68	1.90	0.03	3.37	2.71	0.43	0.0	0.0	2.72	1.50	0.50	0.0	0.0	0.0	2.22
FastViT [53]	98.89	14.63	39.08	4.88	33.29	1.12	15.83	11.21	49.76	9.56	65.73	53.46	1.69	48.72	38.53	0.0	20.91	29.84
SHViT-S1 [54]	10.41	2.24	13.02	1.97	12.06	0.91	13.08	5.43	0.63	0.03	0.80	46.59	4.22	2.91	52.11	0.0	3.52	9.99
SwiftFormer [55]	95.31	18.40	42.68	5.24	28.31	2.46	22.58	42.97	52.20	1.13	44.83	88.60	20.33	71.49	90.41	0.88	68.30	40.94
TinyViT-5M [56]	99.92	92.73	91.47	48.63	66.49	40.27	60.58	47.58	94.30	97.03	97.76	97.52	96.97	99.08	97.11	86.08	98.77	83.07
MobileNetV2 [49]	6.86	11.15	31.18	1.01	13.22	0.65	22.71	1.42	0.06	0.0	0.36	2.55	2.36	8.95	0.0	0.0	0.02	6.02
MobileNetV3 [50]	53.86	0.30	4.94	0.31	5.46	0.75	4.35	12.43	0.96	0.0	0.70	8.78	0.13	1.58	17.22	0.0	1.30	6.65
GhostNet [51]	6.92	2.61	3.11	0.32	12.92	0.55	10.62	7.53	0.03	0.0	0.13	4.08	4.02	5.58	0.0	0.0	0.36	3.45
RepGhostFAS (Student)	0.0	0.05	0.10	0.06	3.94	0.03	2.76	1.64	0.0	0.0	0.0	2.86	4.61	13.20	0.0	0.0	0.13	1.72

AMY: Combined Datasets.

[55], TinyViT-5M [56]) fail to deliver a proportional gain in performance despite their relatively high computational complexity. MobileNetV2 [49], GhostNet [51], and MobileNetV3 [50] provide the optimal balance between performance and efficiency, positioning them as competitive lightweight models. In contrast, *RepGhostFAS* 1 $\times$ achieves the lowest HTER with only 2.79M parameters and 0.38G FLOPs, demonstrating the most favorable balance between efficiency and performance among all compared models.

Table XVI further evaluates computational cost on CPU, GPU, and the Edge controller. While some lightweight models achieve competitive latency, they typically consume more memory or show weaker generalization. The *RepGhostFAS* (Teacher) improves performance with reasonable latency and memory usage, and the distilled *RepGhostFAS* (Student) further reduces computational cost while outperforming other models, delivering the best overall performance.

I. Adaptation to Edge/IoT Deployment

To validate the practical applicability of the proposed *RepGhostFAS* (Student) in edge/IoT environments, we illustrate the complete model optimization and deployment workflow in Fig. 9. The workflow bridges offline training and real-time edge inference, demonstrating how the model is prepared for execution on resource-constrained devices.

The pipeline consists of four stages. First, the *RepGhostFAS* (Student) is trained on FP32 precision. Second, the trained model undergoes a model optimization phase to improve performance and efficiency. This includes Post-Training Quantization (PTQ), where the FP32 model is converted to lower precision (INT8), significantly reducing model size, memory usage, and computational cost. To recover the performance degradation caused by quantization, a fine-tuning step is applied as shown in Table XVII. Third, the optimized model is compiled using OpenVINO [109] into an edge-compatible Intermediate Representation (IR) format (i.e., .xml/.bin), enabling efficient execution on heterogeneous hardware. Lastly, the final adapted model is deployed on a Raspberry Pi 4, where inference is performed at an input resolution of 256 $\times$ 256, achieving 4.45 FPS with an energy consumption of 3.44

J/frame and a 1.49 $\times$ speedup, with only a marginal increase of 0.25% and 0.11% in HTER and EER, respectively.

VI. DISCUSSION

The experimental results presented in this work highlight several key observations regarding the effectiveness of the proposed *RepGhostFAS* framework. First, the intra-dataset results show that *RepGhostFAS* achieves very high accuracy, indicating strong discriminative capability across both simple and complex attack types. Second, comparisons with recent state-of-the-art methods confirm that *RepGhostFAS* delivers competitive or superior performance while maintaining a low computational footprint. Third, the t-SNE and Grad-CAM visualizations show that the model learns discriminative and interpretable representations, forming well-separated feature clusters and focusing on meaningful facial regions. Fourth, the proposed ensemble-based feature fusion strategies effectively mitigate domain shift by leveraging complementary information from multiple domains. In particular, the summation-based fusion achieves the best performance and is adopted as the *RepGhostFAS* (Teacher) model, whose knowledge is transferred to a compact *RepGhostFAS* (Student) model via distillation, significantly reducing computational cost while preserving generalization capability. Fifth, ablation studies on network scaling and input resolution indicate that increasing model width or spatial resolution does not yield proportional performance gains. The *RepGhostFAS* 1 $\times$ configuration with 256 $\times$ 256 input resolution provides the best trade-off between performance and efficiency. In addition, the limitations of both *RepGhostFAS* (Teacher) and *RepGhostFAS* (Student) are observed under fine-grained, hyperparameters, and loss terms variations. Sixth, cross-dataset evaluation demonstrates that *RepGhostFAS* (Student) exhibits strong robustness to unseen attack types, outperforming existing lightweight approaches. Seventh, the computational complexity analysis further confirms that *RepGhostFAS* achieves remarkable performance with reduced computational cost. Finally, the optimized *RepGhostFAS* (Student) model is deployed on a constrained IoT device (Raspberry Pi 4), demonstrating its practical applicability in real-world scenarios.

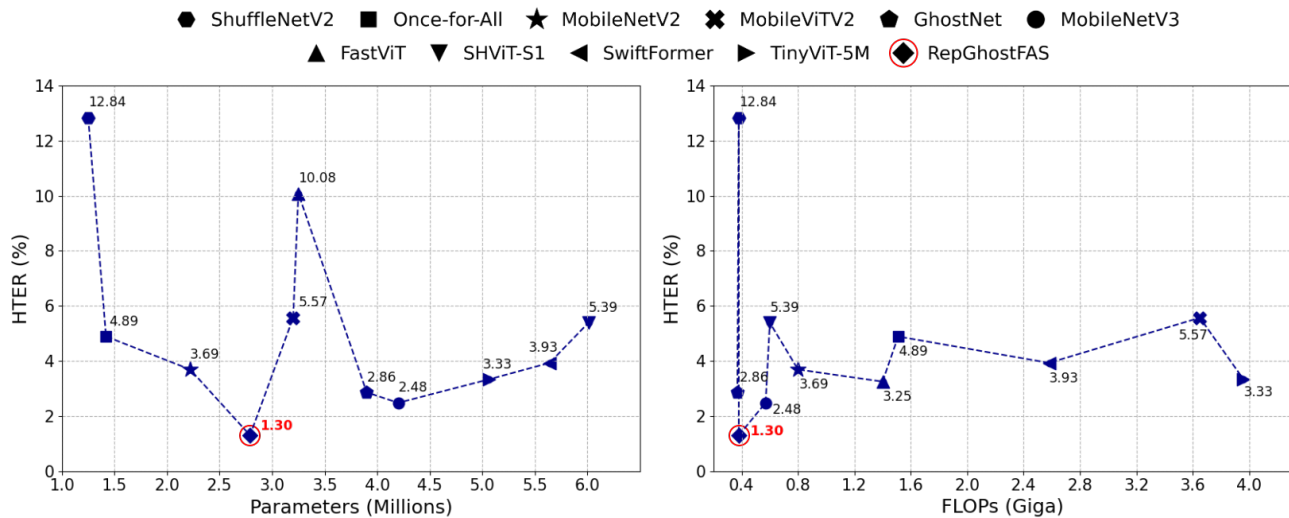


Fig. 8. Performance comparison of lightweight models on the ROSE-Youtu dataset (Left: HTER vs. parameters, Right: HTER vs. FLOPs). The proposed RepGhostFAS achieves the lowest HTER with minimal computational cost.

TABLE XVI
PERFORMANCE VS. COMPUTATIONAL COMPLEXITY COMPARISON.

Train (AMY)	Test	Individual-Dataset		Cross-Dataset	Computation Cost				
		Avg. (A & M & Y)		Avg. (All Attack Types)	Resolution (256 × 256)				
		HTER (%)	EER (%)	EER (%)	CPU	GPU		Controller	
					Latency (ms)	Latency (ms)	Memory (MB)	Latency (ms)	Memory (MB)
ShuffleNetV2 [47]		0.82	0.75	2.65	22.24	10.75	356	24.10	17.80
Once-for-All [48]		0.35	0.20	3.01	44.46	16.35	376	36.48	47.50
MobileViTV2 [52]		0.63	0.55	2.22	34.45	10.82	498	27.12	48.71
FastViT [53]		3.54	2.30	29.84	38.08	14.47	362	35.10	34.51
SHViT-S1 [54]		4.63	4.26	9.99	20.06	16.07	976	40.43	35.05
SwiftFormer [55]		3.70	2.04	40.94	39.13	15.02	382	36.51	40.52
TinyViT-5M [56]		9.20	9.04	83.07	37.41	14.17	390	31.12	62.54
MobileNetV2 [49]		2.74	2.22	6.02	28.99	8.98	360	20.19	36.50
MobileNetV3 [50]		4.20	2.16	6.65	28.12	9.93	366	23.92	48.71
GhostNet [51]		2.93	0.87	3.45	36.17	16.98	366	33.57	32.99
RepGhostFAS (Baseline)		2.36	1.12	2.90	30.50	14.26	362	36.45	30.03
RepGhostFAS Ensemble (Concatenation)		0.86	0.91	4.18	107.02	45.35	490	110.88	36.50
RepGhostFAS Ensemble (Summation) §		0.47	0.38	3.52	105.12	43.40	488	111.27	31.54
RepGhostFAS Ensemble (Attention)		0.61	0.43	3.50	108.74	45.57	490	111.91	39.67
RepGhostFAS (Student)		0.77	0.49	1.72	30.50	14.26	362	36.45	30.03

AMY: Combined Datasets, All Attack Types: SynthASpoof & SOTERIA & UniAttackData, § RepGhostFAS (Teacher), CPU: Intel® Core™ i5-10400@2.90GHz, GPU: NVIDIA GeForce RTX 2080 Ti, Controller: NVIDIA Jetson Orin Nano.

Furthermore, we emphasize that our evaluation is conducted under a unified image-based protocol. While temporal models can leverage motion cues, our goal is to ensure efficiency under strict computational constraints. To partially bridge this gap, we extend the RepGhostFAS (Student) model to a simple multi-frame inference setting by aggregating predictions across frames. As expected, performance improves with additional frames on the ROSE-Youtu (Y) dataset, reducing HTER from 1.63% to 1.38% and EER from 1.09% to 0.97% when using 10 frames.

Future work may explore extending RepGhostFAS as a backbone for multimodal FAS systems that integrate depth, thermal, infrared, remote photoplethysmography (rPPG), and

optical flow modalities, as well as temporal information from video sequences. While such approaches can further improve performance under challenging conditions, they require additional sensing hardware and increased computational resources.

Overall, RepGhostFAS provides a practical and efficient solution for lightweight face anti-spoofing with strong robustness and low computational cost, demonstrating its technical feasibility for deployment on IoT and edge devices. Nevertheless, additional evaluation of demographic fairness, broader deployment conditions, and subgroup-specific failure modes is required before its use in sensitive real-world biometric authentication systems.

TABLE XVII
PERFORMANCE AND EFFICIENCY COMPARISON OF *RepGhostFAS (Student)*, SHOWING THE IMPACT OF QUANTIZATION AND FINE-TUNING, EVALUATED OVER 10 RUNS (MEAN $\pm$ STD).

Model	Model Size (MB)	Dataset (Train $\rightarrow$ Test)						Avg.		OpenVINO IR (.xml/bin)	
		AMY $\rightarrow$ A		AMY $\rightarrow$ M		AMY $\rightarrow$ Y				Resolution (256 $\times$ 256)	
		HTER (%)	EER (%)	HTER (%)	EER (%)	HTER (%)	EER (%)	HTER (%)	EER (%)	Latency (ms) CPU	Latency (ms) SBC
RepGhostFAS (Student)	32.72	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	1.63 $\pm$ 0.22	1.09 $\pm$ 0.12	0.54 $\pm$ 0.94	0.36 $\pm$ 0.63	4.90 $\pm$ 0.47	334.42 $\pm$ 29.79
RepGhostFAS (Student) $\dagger$	13.60	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	2.90 $\pm$ 0.24	1.73 $\pm$ 0.30	0.96 $\pm$ 1.67	0.57 $\pm$ 1.00	4.15 $\pm$ 0.13	224.91 $\pm$ 1.16
RepGhostFAS (Student) $\ddagger$	13.60	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	2.39 $\pm$ 0.24	1.41 $\pm$ 0.14	0.79 $\pm$ 1.38	0.47 $\pm$ 0.81	4.15 $\pm$ 0.13	224.91 $\pm$ 1.16

$\dagger$ Quantized (INT8), $\ddagger$ Quantized (INT8) & Fine-tune, AMY: Combined Datasets, CPU: Intel[®] Core[™] i5-10400@2.90GHz, SBC: Raspberry Pi 4 (15.3W).

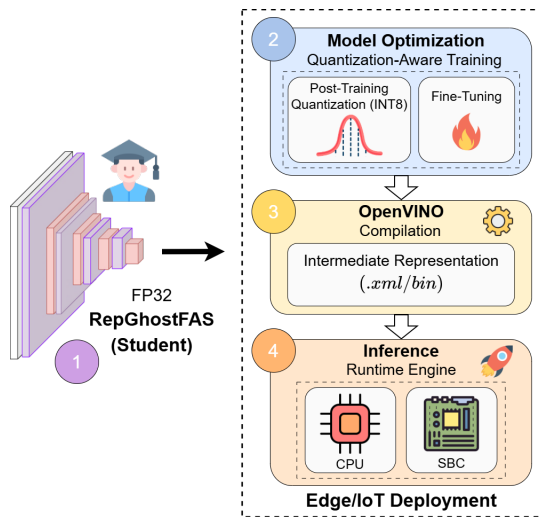


Fig. 9. Deployment pipeline of *RepGhostFAS (Student)* for edge/IoT environments, including quantization, compilation, and inference (batch size = 1).

VII. CONCLUSION

In this paper, we presented *RepGhostFAS*, a lightweight and efficient solution for face anti-spoofing (FAS). To the best of our knowledge, this is the first study to investigate the effectiveness of the RepGhost architecture for FAS, and our results show that it is among the most efficient models compared to existing lightweight approaches. We addressed two major challenges in FAS: domain shift and computational efficiency. To mitigate domain shift, we proposed ensemble-based feature fusion strategies that learn complementary cross-domain representations. Furthermore, to maintain a compact architecture, we employed a knowledge distillation that transfers knowledge from an ensemble teacher model to a single student model. Overall, *RepGhostFAS* offers a cost-effective, resource-efficient, and robust defense against a wide range of spoofing attacks, making it well suited for real-time deployment on IoT and edge devices.

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Taha Hasan Masood Siddique (Graduate Student Member, IEEE) received the B.E. degree in Electronic Engineering from Iqra University, Karachi, Pakistan, in 2016, and the M.E. degree in Electronic Engineering from the NED University of Engineering & Technology, Karachi, in 2020. He is currently pursuing a Ph.D. degree with the College of Information Science & Electronic Engineering at Zhejiang University, Hangzhou, China. From 2017 to 2022, he was a Lecturer with the Faculty of Engineering Sciences & Technology, Iqra University,

Karachi. In 2022, he also served as a Visiting Lecturer at the National Textile University, Karachi. During his Master's studies, he was associated as a Research Assistant with the Autonomous Laboratory, NED University of Engineering & Technology, Karachi. His research interests include machine learning, generative AI, and biometric security.



Shujaat Khan is an Assistant Professor in the Department of Computer Engineering and a Fellow at the Saudi Data and AI Authority (SDAIA) and King Fahd University of Petroleum and Minerals (KFUPM) under the SDAIA-KFUPM Joint Research Center for Artificial Intelligence (JRC-AI) at KFUPM, Dhahran, KSA. Prior to joining KFUPM, he was a Senior AI Scientist at Digital Technology & Innovation, Siemens Medical Solutions USA, Inc. He earned his Ph.D. from the Department of Bio and Brain Engineering at the Korea Advanced

Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2022. He was a researcher with Synergistic Bioinformatics (SynBi) and the Bio Imaging, Signal Processing Learning (BISPL) Labs at KAIST. His research interests include machine learning, optimization, inverse problems, and signal processing, with a focus on biomedical and bioinformatics applications.



Muhammad Shahid Jabbar earned his Ph.D. in Computer Engineering from Sungkyunkwan University, Suwon, South Korea, in 2022. He was a Researcher at the H-Lab and the Media System Lab at SKKU. His doctoral research focused on developing modern deep learning solutions using multimodal data for convergence applications involving multi-sensory interaction and cross-modular associations, with particular emphasis on accessibility for visually impaired users. He is currently a Post-Doctoral Fellow at the SDAIA-KFUPM Joint Research Center

for Artificial Intelligence, King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, KSA. Prior to joining KFUPM, he worked as Senior AI Researcher at Radisen Co. Ltd., Seoul, South Korea, where he contributed to AI-powered digital radiography solutions for medical image analysis. His research interests include computer vision, diffusion models, multimodal learning, and inverse problems, with a focus on biomedical and human-centered artificial intelligence.



Akash Kumar (Senior Member, IEEE) received the Joint Ph.D. degree in Electrical Engineering and Embedded Systems from the Eindhoven University of Technology, Eindhoven, The Netherlands, and the National University of Singapore (NUS), Singapore, in 2009. From 2009 to 2015, he was with NUS. From October 2015 to March 2024, he was a Professor with Technische Universität Dresden, Dresden, Germany, where he directed the Chair for Processor Design. Since April 2024, he has been directing the Chair of Embedded Systems at Ruhr-Universität

Bochum, Bochum, Germany. His research interests include the design and analysis of low-power embedded multiprocessor systems and the development of secure systems using emerging nanotechnologies.



Kejie Huang (Senior Member, IEEE) received the Ph.D. degree from the National University of Singapore, Singapore, in 2014. He has been a Principal Investigator at the College of Information Science & Electronic Engineering, Zhejiang University (ZJU), since 2016. Prior to joining ZJU, he spent five years at Samsung and Xilinx, two years at A*STAR, and another three years at the Singapore University of Technology and Design (SUTD). From 2020 to 2023, he served as an Associate Editor of *IEEE Transactions on Circuits and Systems II*. In 2022–

2023, he was a Guest Editor of *Applied Science*. He has led more than 20 scientific research projects and authored or co-authored over 100 peer-reviewed journal and conference papers. He holds more than 30 granted patents, with an additional 40 patents pending. His awards include the First Prize of China Business Federation for Innovation and Technology in the Service Industry (2023), the Third Prize of CCF for Technological Progress (2024), the Excellent Cooperation Project Award of Zhejiang University–VIVO (2024), the Best Paper Award of CVPR Workshop (2023), the winner of CVPR RVISU Competition (2021), the Huawei Ascend Intelligence Quality Award (2021), the First Prize and Best Instructor Award of the Fifth China Graduate Student Innovative “Chip” Competition (2022), the National–Departmental–Enterprise–Supervision Horizontal Project (2021).